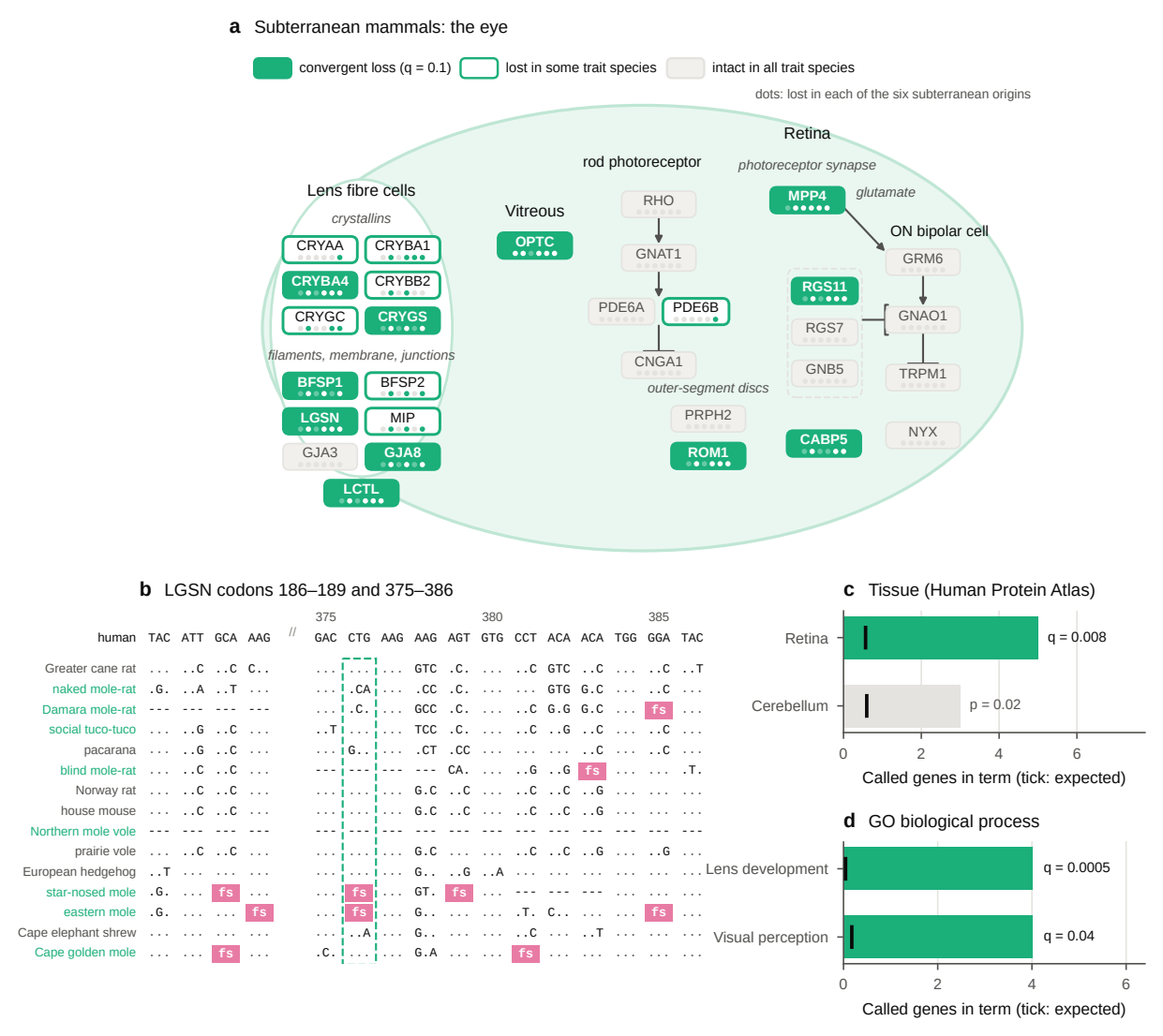

a Subterranean mammals: the eye
convergent loss (q = 0.1)
lost in some trait species
intact in all trait species
dots: lost in each of the six subterranean origins
Retina
rod photoreceptor
photoreceptor synapse
Lens fibre cells
crystallins
Vitreous
RHO
MPP4
glutamate
ON bipolar cell
OPTC
GNAT1
GRM6
CRYAA
CRYBA1
RGS11
CRYBA4
CRYBB2
PDE6A
PDE6B
GNAO1
CRYGC
CRYGS
RGS7
filaments, membrane, junctions
CNGA1
GNB5
TRPM1
BFSP1
BFSP2
outer-segment discs
LGSN
MIP
PRPH2
NYX
GJA3
GJA8
CABP5
ROM1
LCTL
b LGSN codons 186–189 and 375–386
human TAC ATT GCA AAG // GAC CTG AAG AAG AGT GTG CCT ACA ACA TGG GGA TAC
Greater cane rat ... ..C ..C C.. ... ... GTC .C. ... ..C GTC ..C ... ..C ..T
naked mole-rat .G. ..A ..T ... ... .CA ... .CC .C. ... ... GTG G.C ... ..C ...
Damara mole-rat --- --- --- --- ... .C. ... GCC .C. ... ..C G.G G.C ... fs ...
social tuco-tuco ... ..G ..C ... ..T ... ... TCC .C. ... ..C ..G ..C ... ..C ...
pacarana ... ..G ..C ... ... G.. ... .CT .CC ... ...C ... ..C ...
blind mole-rat ... ..C ..C ... --- --- --- --- CA. ... ..G ..G fs ... ... .T.
Norway rat ... ..C ..C ... ... G.C ..C ... ..C ..C ..G ... ...
house mouse ... ..C ..C ... ... G.C ..C ... ..C ..C ..G ... ...
Northern mole vole --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- ---
prairie vole ... ..C ..C ... ... G.C ... .C ..C ..G ... ..G ...
European hedgehog ..T ... ... G.. ..G ..A ... ...
star-nosed mole .G. ... fs ... ... fs ... GT. fs ... --- --- --- ... ...
eastern mole .G. ... ... fs ... fs ... G.. ... ... .T. C.. ... ... fs ...
Cape elephant shrew ... ...A ... G.. ... .C ... ..T ... ...
Cape golden mole ... ... fs ... .C. ... ... G.A ... ... fs ... ...
c Tissue (Human Protein Atlas)
Retina
q = 0.008
Cerebellum
p = 0.02
0
2
4
6
Called genes in term (tick: expected)
d GO biological process
Lens development
q = 0.0005
Visual perception
q = 0.04
0
2
4
6
Called genes in term (tick: expected)